

Qns	Answer	Marks
3(a)(i)	$R_{total} = R_{//} + r$ $= \left(\frac{1}{600} + \frac{1}{3000} \right)^{-1} + 30 = 530 \, \Omega$	M1
	$V_{LDR} = V_{terminal} = \mathcal{E} - Ir$ $= 12 - \left(\frac{12}{530} \right)(30) = 11.3 \, \text{V}$	M1
	$I_{LDR} = \frac{V_{LDR}}{R_{LDR}}$ $= \frac{11.3}{3000} = 3.77 \, \text{mA} = 3.8 \, \text{mA} \, (2\text{sf}) \, (\text{shown})$	M1
OR		
$R_{//} = \left(\frac{1}{600} + \frac{1}{3000} \right)^{-1} = 500 \, \Omega$		
$V_{LDR} = V_{//} = \frac{R_{//}}{R_{//} + r} \times \mathcal{E}$ $= \frac{500}{500 + 30} \times 12 = 11.3 \, \text{V}$		
$I_{LDR} = \frac{V_{LDR}}{R_{LDR}}$ $= \frac{11.3}{3000} = 3.77 \, \text{mA} = 3.8 \, \text{mA} \, (2\text{sf}) \, (\text{shown})$		
OR		
$R_{total} = R_{//} + r$ $= \left(\frac{1}{600} + \frac{1}{3000} \right)^{-1} + 30 = 530 \, \Omega$		

	$I_{Tot} = \frac{\varepsilon}{R_{Tot}} = \frac{12}{530} = 0.0226 \text{ A}$ $I_{LDR} = \frac{600}{600 + 3000} \times I_{Tot}$ $= \frac{600}{600 + 3000} \times 0.0226 = 3.77 \text{ mA} = 3.8 \text{ mA (2sf) (shown)}$	
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(a)(ii)	$P_{\text{LDR}} = I_{\text{LDR}}^2 R_{\text{LDR}}$ $= (3.8 \times 10^{-3})^2 (3000)$ $= 0.043 \text{ W (2sf)}$	A1
(b)	<u>Explanation 1 (preferred):</u> Resistance of LDR decreases in bright light, thus effective total resistance of entire circuit decreases. Total current I_{Tot} in circuit increases, resulting in larger potential drop V_r across the internal resistance r Since $V_{\text{terminal}} = \mathcal{E} - V_r = \mathcal{E} - I_{\text{Tot}}r$, where V_{terminal} is the terminal p.d., $\mathcal{E}$ is the e.m.f. of the cell; thus the terminal p.d. decreases. OR <u>Explanation 2:</u> Resistance of LDR decreases in bright light, thus resistance of the combination of parallel resistors decreases. By potential divider rule, potential difference (p.d.) across parallel resistors decreases. Since terminal p.d. is the same as p.d. across parallel resistor, terminal p.d. decreases.	B1 B1 B1

Qns	Answer	Marks
4(a)	Magnetic flux density B is the force acting per unit current per unit length on a	B1